

Coding Set 5

Q1. Student Information System

Create a base class Student containing name, roll number, and age. Derive a class EngineeringStudent containing branch and semester. Use constructors to initialize all data members and display the complete details of the student.

Q2. Employee Management System

Create a base class Employee containing employee ID and name. Derive a class Manager containing department and salary. Create an array of 5 manager objects, initialize them using constructors, and display their details.

Q3. Library Management System

Create a base class Book containing title and author name. Derive a class EBook containing file size and file format. Store details of three e-books in an array of objects and display all information.

Q4. Vehicle Registration System

Create a base class Vehicle containing registration number and company name. Derive classes Car and Bike containing additional attributes such as fuel type and engine capacity. Create multiple objects and display their details.

Q5. Banking System Using Function Overriding

Create a base class Account containing account number and balance. Derive classes SavingsAccount and CurrentAccount. Override the display() function in the derived classes to show account-specific details. Demonstrate the concept using multiple objects.

Q6. Template Function for Generic Operations

Write template functions to:

- Find the larger of two values
- Swap two values

Demonstrate the functions using int, float, double, and char data types.

Q7. Template Class for Pair of Values

Create a template class `Pair<T>` that stores two values of the same type. Implement functions to find and display the maximum and minimum values. Test the class with integers and floating-point numbers.

Q8. Generic Array Class

Create a template class `Array<T>` that stores five elements of any data type. Implement functions to:

- Input elements
- Display elements
- Find the largest element
- Find the smallest element

Demonstrate the class using integer and floating-point arrays.

Q9. Student Result Processing System

Create a template class `Result<T>` to store marks of five subjects. Implement member functions to calculate:

- Total marks
- Average marks
- Highest marks
- Lowest marks

Display the complete result of a student using both integer and floating-point marks.

Q10. University Record Management System

Create a base class `Person` containing name and age. Derive classes `Teacher` and `ResearchScholar` with appropriate additional attributes. Create a template class `RecordManager<T>` capable of storing and displaying records of any type. Use it to manage teacher and research scholar records.